Supporting Information S1

Offline paired evaluation of MRFoundry 0.14.0

This supplement reports saved offline comparisons of the same agent with and without MRFoundry 0.14.0 (commit bd49b95). A paired trial consists of one attempt per arm with matched model, tools, inputs and nominal budgets, subject to the resource-policy caveats in S7.4. Verified completion requires every automated grader check to pass. This document introduces no new model execution or grading.

Study design

Core includes one sealed instance of each of 24 task types for ten models: 240 paired trials. Hard contains 15 task types: 11 multi-stage and four system-design types. Three models cover all 15 types with two instances each (90 pairs); four further models cover the 11 multi-stage types with one instance each (44 pairs). The resulting 134 Hard pairs are not a uniform model leaderboard. Hard task identities and materials remain private.

Table S1. Aggregate completion and paired effects

Set	Pairs	Without -> with	Change, pp	95% CI, pp	P
Core	240	162/240 -> 188/240	+10.8	[+0.8, +21.2]	0.0004
Hard	134	49/134 -> 60/134	+8.2	[-3.9, +19.7]	0.061

Intervals are the saved 95% two-way cluster bootstrap over task types and models (4,000 resamples, seed 0). P values are saved secondary, unclustered exact two-sided McNemar comparisons, which assume independent discordant-pair signs and do not adjust for task/model clustering (see S7.3). Core discordant pairs are 39 with-only and 13 without-only; Hard discordant pairs are 20 and 9. The Hard cluster-bootstrap interval includes zero.

Scope and recorded limitations

- No web search was available in the offline evaluation. Only Core and Hard are reported; internal later-round data are excluded.

- One additional model was excluded after its API rejected the host file-editing tool; nine started pairs are outside the analysis.

- A runner setup failure affected 143 pair setups before either arm reached a model turn (94 Core, 49 Hard). These were restarted without changing the task or model condition.

- An 8 GiB per-process memory cap began partway through the runs. Recorded settings matched the cap within newly prepared pairs; exposure records are incomplete and resources were not uniform across the complete study. S7.4 details the policy changes and documentation limits.

- One Hard task has an image subcheck that accepts a zero-filled reconstruction on one instance. Its identity is withheld; the aggregate scores remain the originally recorded scores.

Model and domain summaries below are descriptive. Their different coverage and the exploratory breakdowns do not establish a general model ranking or corrected multiple-comparison claims.

Core: full matched coverage

24 task types per model; 240 paired trials

Table S2. Core completion by domain

Domain	Pairs	Without -> with	With-only / without-only
Diffusion	20	17/20 -> 15/20	0 / 2
Functional MRI	10	8/10 -> 9/10	1 / 0
Hardware / field	20	13/20 -> 14/20	3 / 2
Perfusion / CEST	20	14/20 -> 18/20	4 / 0
Reconstruction	20	13/20 -> 15/20	4 / 2
Relaxometry / QSM	20	13/20 -> 19/20	6 / 0
Sequence design	70	34/70 -> 45/70	14 / 3
Spectroscopy	10	8/10 -> 8/10	1 / 1
Study design / statistics	20	17/20 -> 17/20	2 / 2
Writing / review	30	25/30 -> 28/30	4 / 1

Diffusion decreased from 17 to 15 passes across 20 paired trials. Domain denominators differ because each domain contains a different number of task types.

Table S3. Core completion by model (alphabetical)

Model	Pairs	Without -> with	With-only / without-only
Claude Haiku 5.5	24	15/24 -> 19/24	7 / 3
Claude Opus 5.5	24	22/24 -> 22/24	1 / 1
Claude Sonnet 5.5	24	17/24 -> 23/24	6 / 0
Gemini 3.8 Flash	24	19/24 -> 22/24	4 / 1
Gemini 4 Argon	24	21/24 -> 24/24	3 / 0
GPT-5.3 Codex	24	9/24 -> 13/24	5 / 1
GPT-6.1 Sol	24	19/24 -> 22/24	3 / 0
GPT-6.1 Terra	24	22/24 -> 22/24	2 / 2
Kimi K3	24	13/24 -> 16/24	6 / 3
MAI Code 1.1 Flash	24	5/24 -> 5/24	2 / 2

All models have identical task-type coverage in Core. Seven models improved in pass count, three were unchanged, and none declined in aggregate. These totals can conceal task-specific gains and losses, which are retained in the discordant-pair counts.

Hard: separate coverage cohorts

Different task coverage is reported explicitly

Table S4a. Full coverage: 15 task types, two instances each

Three models; 30 paired trials per model. Cohort total: 44/90 without skills and 52/90 with MRFoundry.

Model	Pairs	Without -> with	With-only / without-only
Claude Opus 5.5	30	16/30 -> 20/30	5 / 1
Gemini 4 Argon	30	13/30 -> 14/30	5 / 4
GPT-6.1 Sol	30	15/30 -> 18/30	6 / 3

Table S4b. Multi-stage coverage: 11 types, one instance each

Four models; 11 paired trials per model. Cohort total: 5/44 without skills and 8/44 with MRFoundry. Models are alphabetical within each cohort, not ranked across cohorts.

Model	Pairs	Without -> with	With-only / without-only
Claude Haiku 5.5	11	2/11 -> 3/11	2 / 1
Gemini 3.8 Flash	11	2/11 -> 4/11	2 / 0
GPT-5.3 Codex	11	1/11 -> 1/11	0 / 0
MAI Code 1.1 Flash	11	0/11 -> 0/11	0 / 0

Table S5. Hard completion by task class

Task class	Pairs	Without -> with	With-only / without-only
Multi-stage (11 types)	110	48/110 -> 57/110	18 / 9
System design (4 types)	24	1/24 -> 3/24	2 / 0

Table S6. Reported token use (with / without)

Set	Total tokens ratio	Tokens per verified success
Core	1.048	0.903
Hard	1.062	0.868

Tokens per verified success divides all reported tokens, including unsuccessful attempts, by the number of verified successes. It is not a price or billing estimate; token accounting can differ between model providers.

Frozen source provenance

All displayed counts are checked against the saved paired records. Intervals and token ratios are copied from the saved summary, not recomputed. The shareable figure-5-data.json contains only approved aggregates; it contains no Hard task IDs or per-task outcomes.

offline-summary.json SHA-256: dfadc34f3fa62642e59ec4c0f418e1af3b083c79b1aa38c3983e19ddae029437

offline-pairs.json SHA-256: 86d2320eef791d4a3bd9fa310f886edba435f8f51bde69bafbed808a59b67760